Answer **all** the questions in the spaces provided.

1 A smooth pebble, made from uniform rock, has the shape of an elongated sphere as shown in Fig. 1.1.

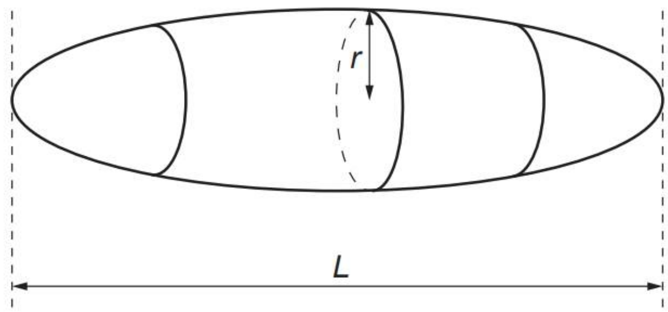

Fig. 1.1

The length and the mass of the pebble is L and M respectively. The cross-section of the pebble, in the plane perpendicular to L, is circular with a maximum radius r.

The density ρ of the rock from which the pebble is composed is given by

$$\rho = \frac{Mr^n}{kL}$$

where n is an integer and k is a constant, with no units, that is equal to 2.094.

(a) Use SI base units to show that n is equal to −2.

[1]

(b) The mean values, absolute uncertainties and percentage uncertainties of some of the above quantities are listed in Fig. 1.2.

quantity	mean value	absolute uncertainty	percentage uncertainty
L	0.1242 m	0.0002 m	-
M	1072 g	-	0.18 %
ρ	2340 kg m^{-3}	50 kg m^{-3}	-

Fig. 1.2

(i) Calculate the fractional uncertainty in r.

fractional uncertainty = ... [2]

(ii) Determine r with its absolute uncertainty. Give your values to the appropriate number of significant figures.

r = (...................... ± ....) m [2]

2 (a) State *Newton's third law of motion*.